Comparison of the Effectiveness of Prompting Techniques (Role-Based) Across Different AI Platforms

Abstract

Prompt engineering has become an important method for improving the quality of responses produced by modern artificial intelligence systems. Large language models such as ChatGPT, Claude, Gemini, and Copilot generate outputs based on the instructions they receive from users. The way a prompt is written can significantly influence the accuracy, clarity, and usefulness of the response. Among the many prompt engineering techniques, role-based prompting has gained attention because it encourages AI models to respond from the perspective of a specific professional role such as a teacher, developer, researcher, or domain expert.

This report explores the effectiveness of role-based prompting across different AI platforms and compares it with other prompting techniques such as zero-shot prompting, few-shot prompting, and chain-of-thought prompting. The study examines how different AI systems interpret role instructions and how those instructions influence the structure, detail, and reasoning of generated responses.

The objective of this document is to analyze the strengths and limitations of role-based prompting and evaluate whether assigning a role improves response quality across various AI models. The discussion includes explanations of prompting strategies, comparisons between platforms, and an evaluation of user experience when interacting with AI tools. The results highlight how prompt design can significantly affect the quality of AI-generated information and demonstrate the importance of prompt engineering in modern AI applications.

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Zero- Shot Prompting

Zero-shot prompting is one of the simplest prompting techniques used with AI models. In this method, the user provides only a direct instruction without including examples or additional guidance. The AI model must rely entirely on its training knowledge to generate an appropriate response.

For example, a user might ask an AI system to explain blockchain technology or describe a scientific concept without providing any sample output. The model analyzes the request and produces a response based on patterns learned during training.

Zero-shot prompting is widely used because it is fast and convenient. Users can obtain information quickly without spending time designing complex prompts. Many AI models are capable of producing useful responses even with minimal instructions.

However, the simplicity of zero-shot prompting can sometimes lead to inconsistent results. Since the AI does not receive examples or role instructions, it may interpret the request in multiple ways. This can affect the level of detail, writing style, or accuracy of the output.

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Zero-shot prompting is one of the simplest prompting techniques used with AI models. In this method, the user provides only a direct instruction without including examples or additional guidance. The AI model must rely entirely on its training knowledge to generate an appropriate response.

For example, a user might ask an AI system to explain blockchain technology or describe a scientific concept without providing any sample output. The model analyzes the request and produces a response based on patterns learned during training.

Zero-shot prompting is widely used because it is fast and convenient. Users can obtain information quickly without spending time designing complex prompts. Many AI models are capable of producing useful responses even with minimal instructions.

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Few- Shot Prompting

Few-shot prompting is a technique where the AI model is given several examples before generating its final response. These examples demonstrate the format and style expected in the output.

For instance, if a user wants the AI to produce explanations in a specific format, the prompt may include two or three example explanations. The model studies these examples and then generates a response for the new task using the same structure.

This method improves consistency because the AI can learn patterns from the examples. Few-shot prompting is particularly useful for tasks such as classification, translation, and structured content generation.

Although few-shot prompting often produces more accurate results than zero-shot prompting, it requires additional effort from the user. The prompt must include well-designed examples that clearly illustrate the expected output.

In many AI platforms, few-shot prompting significantly improves performance because it reduces ambiguity and guides the model more effectively. However, the improvement varies depending on the architecture and training of the AI system being used.

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Chain- of- Thought Prompting

Chain-of-thought prompting encourages AI models to explain their reasoning process step by step. Instead of providing a single final answer, the model breaks down the reasoning into multiple stages.

This technique is particularly useful for tasks that involve logical reasoning, mathematics, or complex problem solving. By presenting intermediate steps, the model produces responses that are easier for users to understand and verify.

Researchers have found that chain-of-thought prompting can significantly improve the accuracy of AI models when solving reasoning tasks. When the model explains its thinking process, it is less likely to produce incorrect conclusions.

However, chain-of-thought prompting may increase the length of the response and slightly reduce the speed of generation. Despite this limitation, it remains one of the most powerful prompt engineering techniques for analytical tasks.

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Role- Based Prompting

Role-based prompting instructs the AI model to respond from the perspective of a specific profession or identity. This approach allows the AI to adopt the tone, knowledge level, and communication style associated with the assigned role.

For example, the prompt might say: “Act as a blockchain expert and explain how blockchain works.” When the model receives this instruction, it attempts to provide a more technical and professional explanation.

Role-based prompting is widely used in education and professional environments. Teachers may ask AI to explain topics in a beginner-friendly way, while developers may request technical explanations suited for advanced learners.

This technique often improves the relevance and clarity of responses because the AI adapts its communication style to match the assigned role. However, the effectiveness of role-based prompting depends on how well the AI model interprets the role instruction.

Different AI platforms may respond differently to the same role-based prompt. Some models strongly adjust their tone and detail level, while others show only minor changes in response style.

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Role-based prompting is widely used in education and professional environments. Teachers may ask AI to explain topics in a beginner-friendly way, while developers may request technical explanations suited for advanced learners.

This technique often improves the relevance and clarity of responses because the AI adapts its communication style to match the assigned role. However, the effectiveness of role-based prompting depends on how well the AI model interprets the role instruction.

Different AI platforms may respond differently to the same role-based prompt. Some models strongly adjust their tone and detail level, while others show only minor changes in response style.

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Comparison of AI Platforms

Several AI platforms are commonly used for natural language generation tasks. These include ChatGPT, Claude, Gemini, and Copilot. Although they are all based on large language models, they differ in architecture, training methods, and design priorities.

ChatGPT is known for its strong instruction-following ability and detailed explanations. It often responds well to role-based prompts because it can adjust tone and depth according to the assigned role.

Claude is designed with a focus on safety and thoughtful responses. It tends to produce structured and cautious explanations. While it supports role-based prompting, the change in response style may be less dramatic than in some other models.

Gemini is optimized for integration with web services and productivity tools. It performs well in educational and informational tasks but may rely more heavily on example-based prompts.

Copilot is commonly used in coding and productivity environments. It performs especially well in technical tasks but may vary in how strongly it adapts to role instructions.

Comparing these platforms helps researchers understand how different models respond to prompt engineering strategies.

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Evaluation and Discussion

The effectiveness of role-based prompting can be evaluated using several criteria including accuracy, clarity, completeness, and user satisfaction. When a model receives a role instruction, it attempts to adjust its response to match the expected expertise level.

Experiments with multiple AI platforms show that role-based prompting often improves the structure and professionalism of responses. For example, when the AI is instructed to act as a teacher, it may provide simpler explanations with examples. When instructed to act as an expert, the response may include more technical terminology.

However, the improvement is not identical across all models. Some AI systems respond strongly to role instructions, while others show only moderate differences compared to zero-shot prompts.

User experience is also an important factor. Many users find role-based prompting helpful because it produces responses that match their specific needs. By adjusting the role, users can request beginner-friendly explanations, technical analysis, or professional

recommendations.

Overall, role-based prompting is a flexible and effective technique that enhances communication between humans and AI systems.

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Conclusion

Prompt engineering has become an essential skill for effectively interacting with modern AI systems. Techniques such as zero-shot prompting, few-shot prompting, chain-of-thought reasoning, and role-based prompting allow users to guide AI models toward producing better outputs.

Among these techniques, role-based prompting stands out because it allows the AI to adapt its communication style according to the assigned role. This improves clarity, relevance, and professionalism in many cases.

The comparison of AI platforms shows that different models respond to role-based instructions in different ways. ChatGPT often demonstrates strong improvements, while Claude, Gemini, and Copilot show varying levels of adaptation.

Understanding how these techniques influence AI responses helps users design better

prompts and obtain more useful results. As artificial intelligence continues to evolve, prompt engineering will remain an important area of research and practical application.

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